



1126

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : **The Spark**

Team Leader Name : **Snehi Patel**

Problem Statement : **AI-Powered Digital Twin of India's Climate using India's National Data (PS-05)**

Team Members & Roles

Team Leader:

Name: **Snehi Patel**

Role: AI/ML Lead & System Architect (Data Pipeline & Temperature Models)

College: L.D. College of Engineering (Ahmedabad, Gujarat)

Team Member-1:

Name: **Juhi Vanjara**

Role: Machine Learning Engineer (Climate Prediction & Simulation)

College: L.D. College of Engineering (Ahmedabad, Gujarat)

Team Member-2:

Name: **Yashvi Sanghvi**

Role: Full-Stack Developer (Dashboard, UI/UX & Model Deployment)

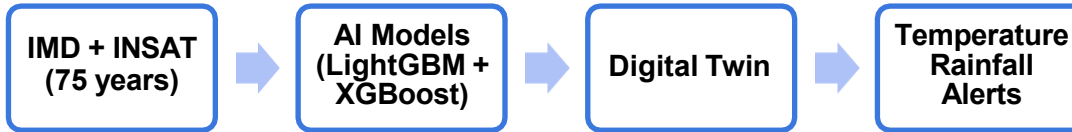
College: L.D. College of Engineering (Ahmedabad, Gujarat)

Opportunity & Key Differentiators:

- How different is it from existing ideas?

Existing	Our Digital Twin
Slow NWP simulations	Instant AI predictions
Large regional forecasts	Grid-level forecasts
Modern data only	Modern AI + Bhadali Knowledge + AI Climate Advisor

- How will it solve the problem?



- USP of the proposed solution:



List of Features Offered by the Digital Twin:

Today: AI-powered Weather Prediction using IMD Rainfall, Temperature, impact assessment, traditional Bhadali features, "what-if" scenario analysis, AI Climate Advisor and INSAT LST datasets.

Next Step: Expand into an AI-powered Climate Prediction Digital Twin by incorporating long-term climate indicators such as CO₂ concentration, greenhouse gases, aerosols, sea surface temperature and enabling climate forecasting

Datasets Ingested:

- IMD Rainfall:

https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html

- IMD Max & Min Temp:

https://imdpune.gov.in/cmpg/Griddata/Max_1_Bin.html

https://www.imdpune.gov.in/cmpg/Griddata/Min_1_Bin.html

- INSAT LST (3RIMG_L2B_LST): <https://www.mosdac.gov.in/>

- INSAT SST & Rainfall (3RIMG_L2B_SST/IMC): <https://www.mosdac.gov.in/> (Will use)



List of Features Offered by the Digital Twin:

1



Daily Climate Forecasting:



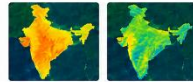
28.6°C

12.4 mm

2



INSAT Satellite Overlay:



LST (°C)

60

30

0

3



Bhadali Moon Phase & Nakshatra Predictor:



Moon Phase



Nakshatra



Heuristic Prediction

4



What-If Scenario Stress Tester:



Temperature Change



Rainfall Change



Sea Level Rise



Impact Assessment



Datasets Ingested:



• IMD Rainfall:

https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html



• IMD Max Temp:

https://imdpune.gov.in/cmpg/Griddata/Max_1_Bin.html



• IMD Min Temp:

https://www.imdpune.gov.in/cmpg/Griddata/Min_1_Bin.html



• INSAT LST (3RIMG_L2B_LST):

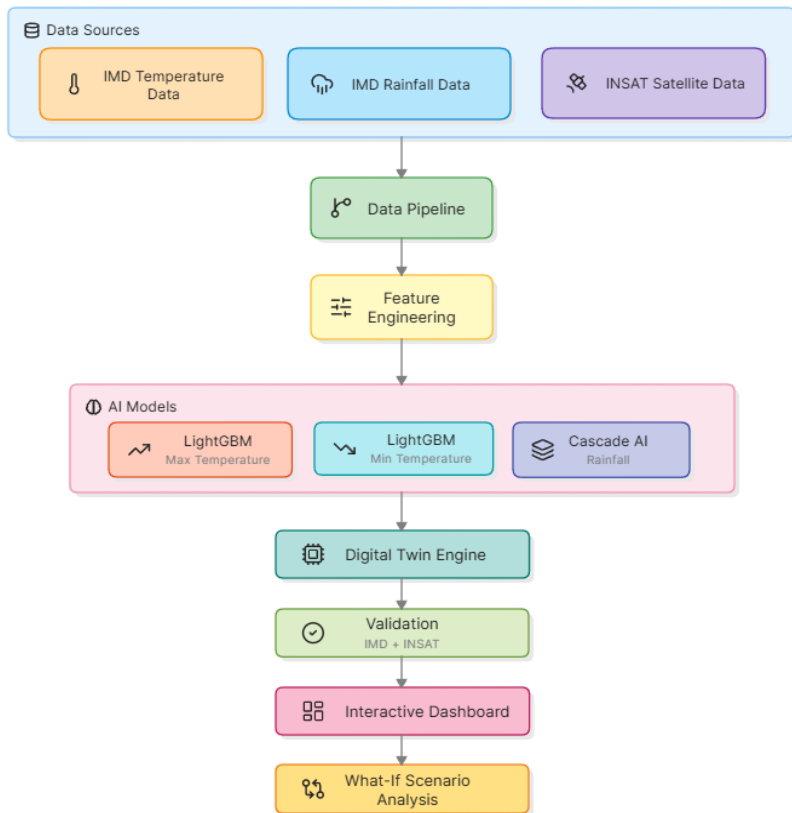
<https://www.mosdac.gov.in/>



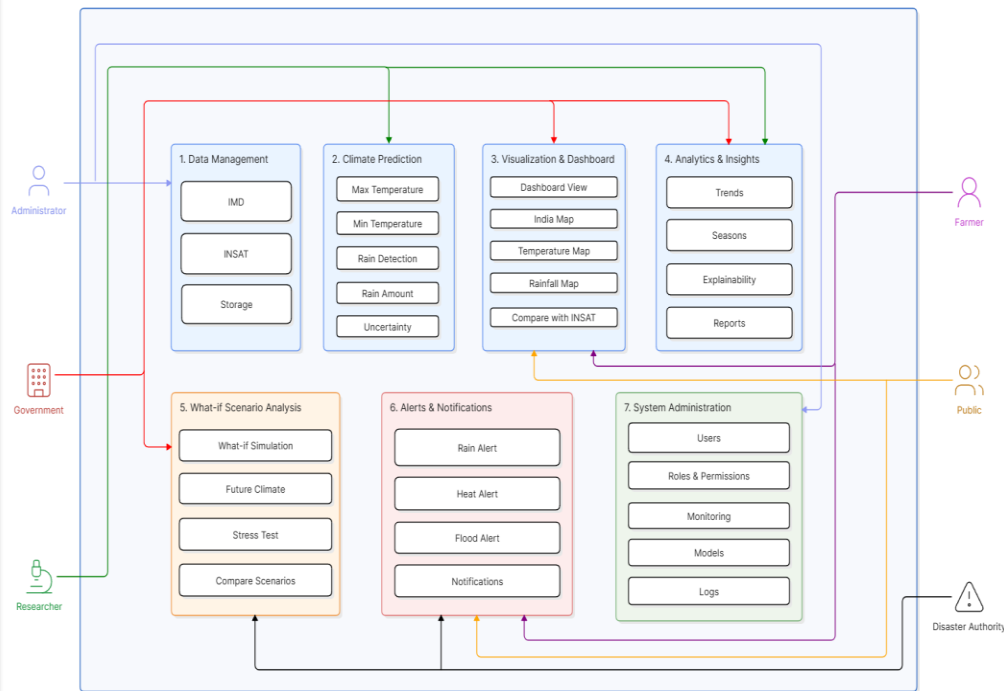
• INSAT SST & Rainfall (3RIMG_L2B_SST/IMC):

<https://www.mosdac.gov.in/>
(Will use)

System Flow Diagram

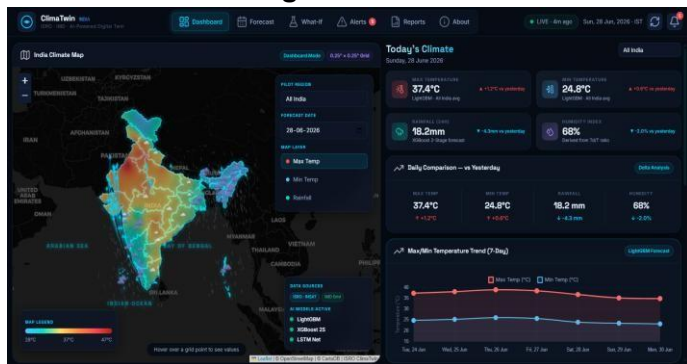


Use Case Diagram

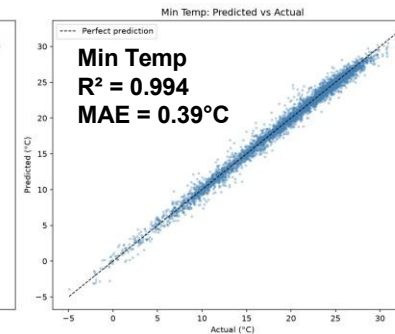
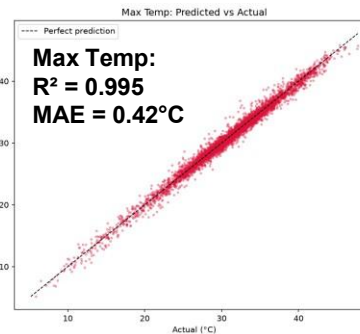
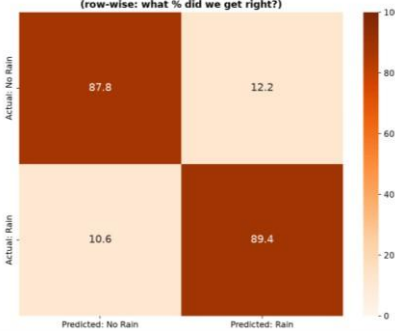
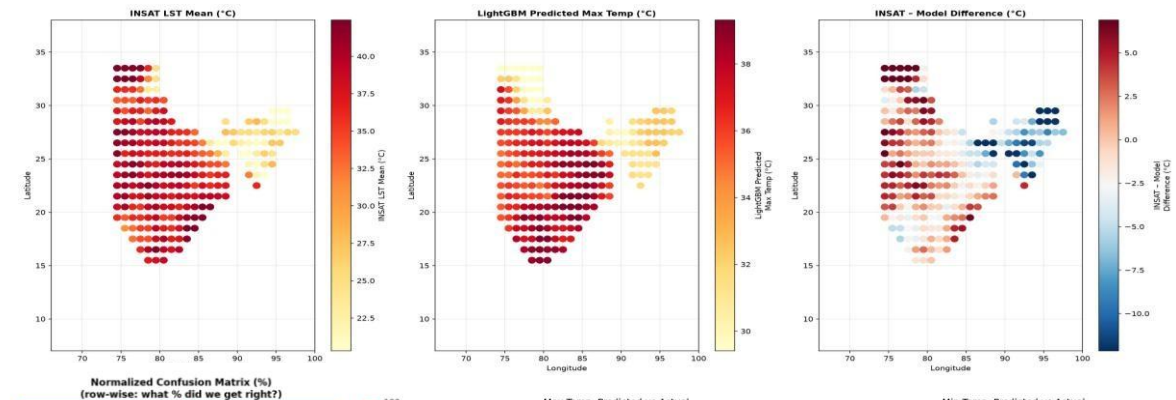


Wireframes/Mock diagrams of the proposed solution

Interactive Digital Twin Dashboard



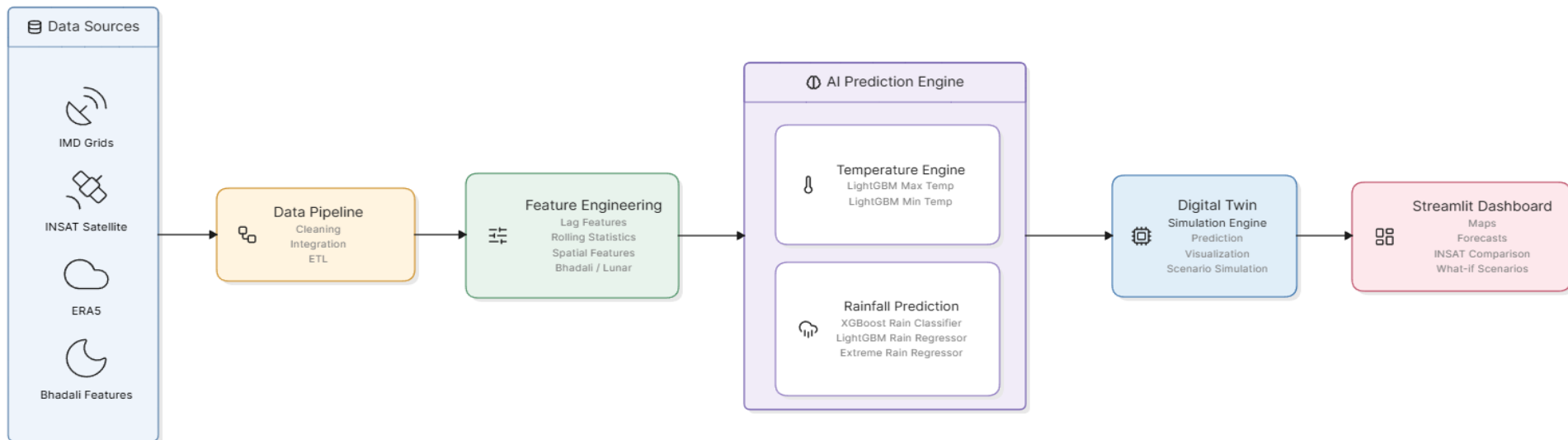
Satellite Validation (INSAT vs AI Prediction)



Rainfall Model Validation
 with ~89.4% accuracy

AI Architecture & Model Accuracies:

- **Max Temperature (LightGBM):** R^2 98.65% | MAE 0.49°C
- **Min Temperature (LightGBM):** R^2 98.92% | MAE 0.49°C
- **Rainfall Detection (XGBoost):** 89.4% Accuracy | F1 0.8765
- Includes **Uncertainty Estimation** (90% Prediction Intervals)
- **Validated Against:** IMD Ground Stations + INSAT Satellite Observations



AI Architecture

Technologies Used in the Proposed Solution:



Estimated Implementation Cost:

Component	Estimated Cost
Software Stack	₹0
IMD & MOSDAC Data	₹0
Cloud Compute	₹3,500–₹6,500/month
Storage	₹500–₹1,000/month
Total Operational Cost	₹4,000–₹7,500/month

Deployment Ready: Docker • AWS • Azure • Google Cloud • On-Premise

BHARATIYA
ANTARIKSH
HACKATHON

2026

THANK YOU